

SHORTEST PATHS WITHOUT A MAP, BUT WITH AN ENTROPIC REGULARIZER

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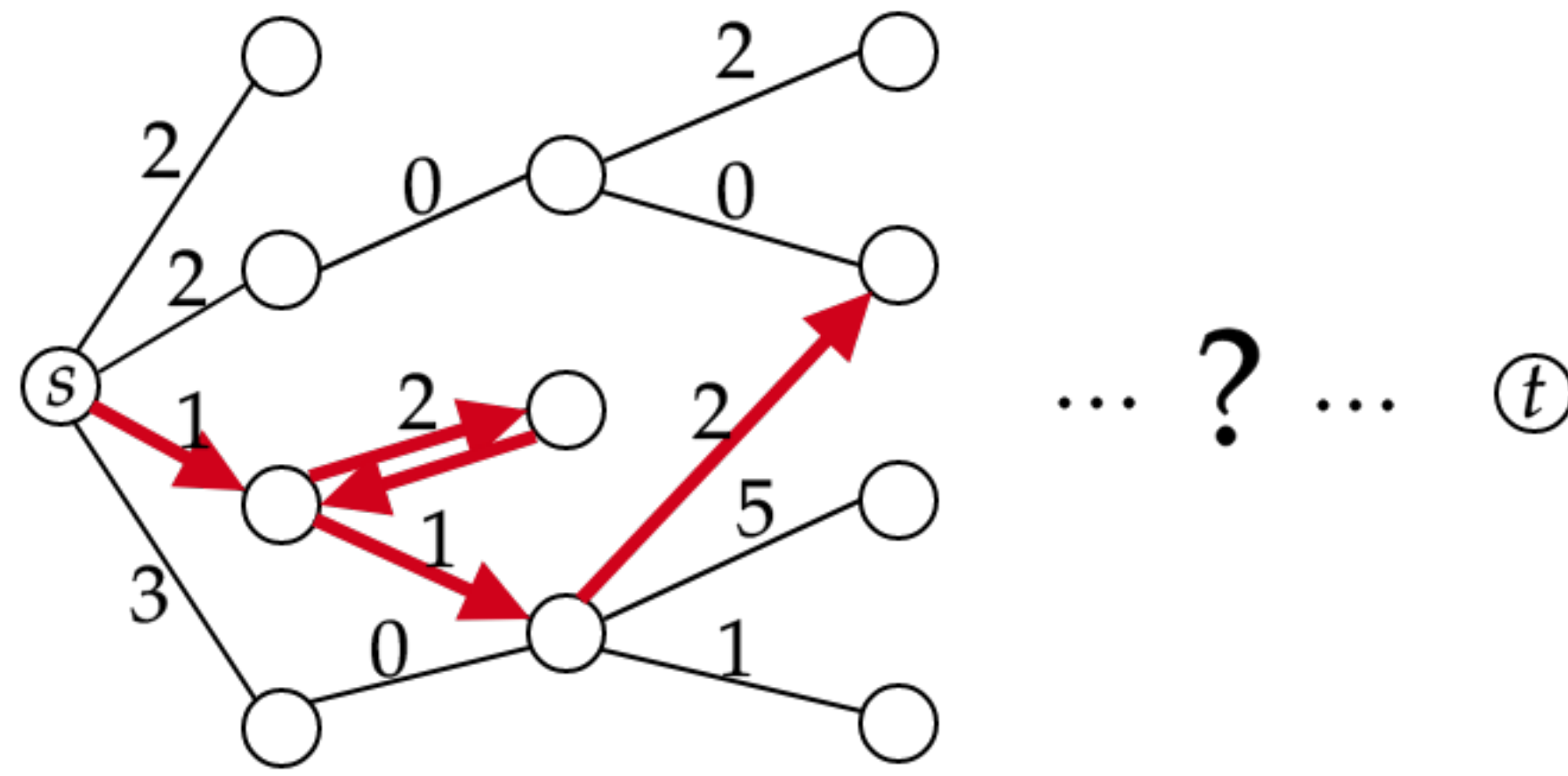
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Layered Graph Traversal [Papadimitriou and Yannakakis, 1989]

Definition

- Edge-weighted graph with vertices in layers $L_0, L_1, \dots, L_n$, edges only between consecutive layers
- Searcher starts at $s \in L_0$. Target $t \in L_n$
- In step i :
 - L_i and edges between L_{i-1}, L_i revealed
 - Searcher goes to vertex in L_i
- Cost = distance traveled until reaching target
- Only parameter: $k = \max_i |L_i|$



State of the art

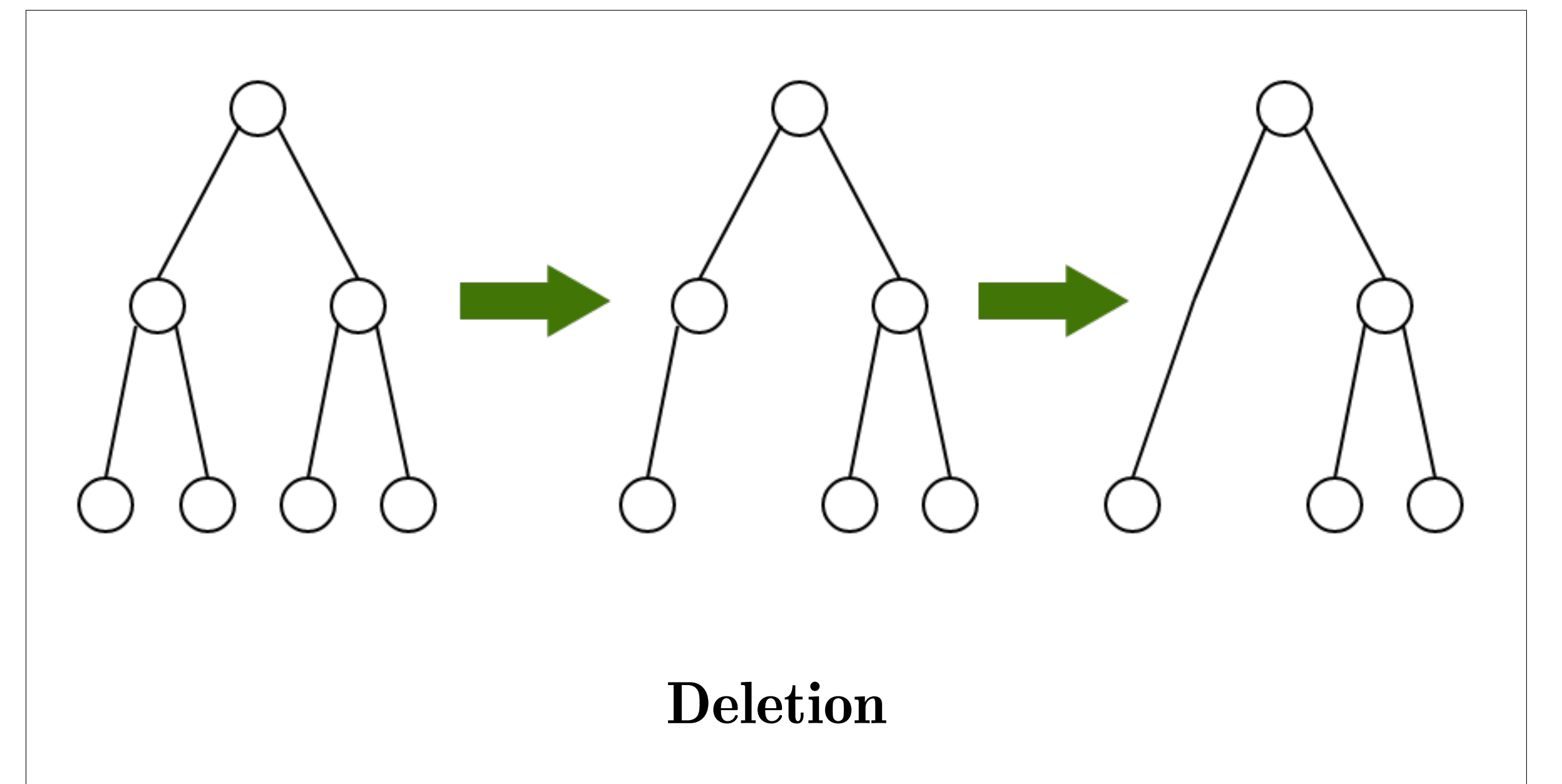
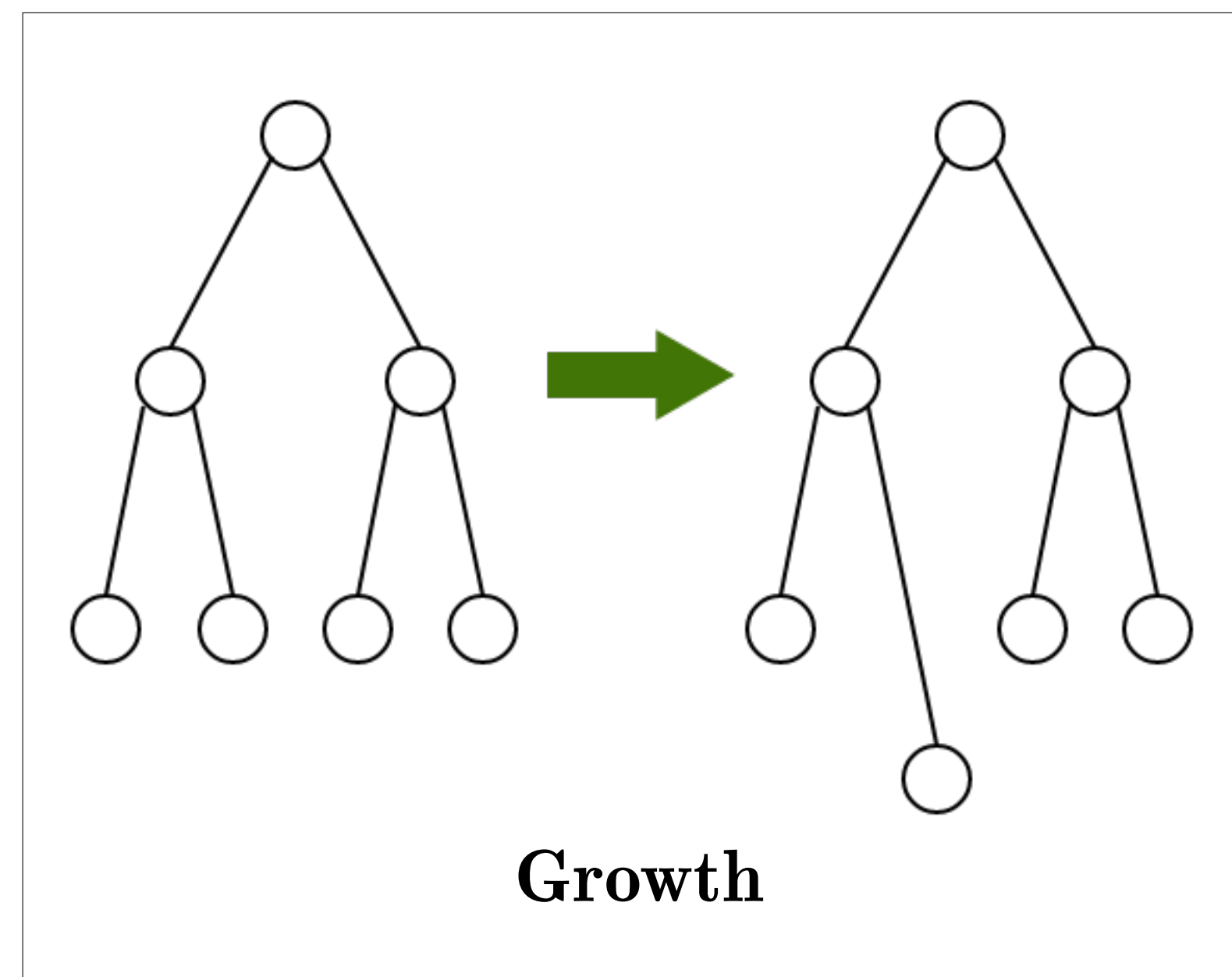
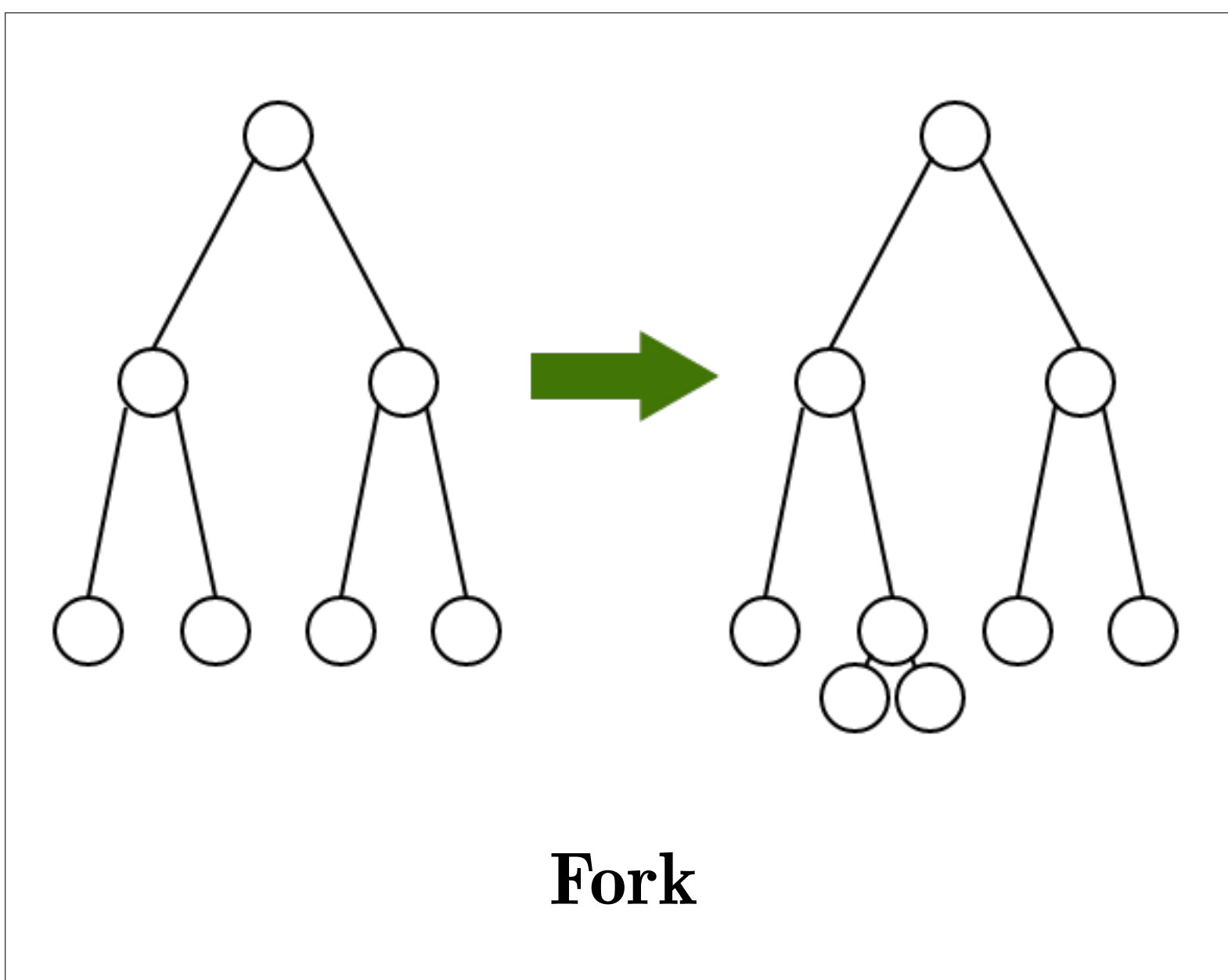
- deterministic: $O(k \cdot 2^k) \cap \Omega(2^k)$ -competitive [Burley, 1996, Fiat et al., 1991]
- randomized: $O(k^{13}) \cap \Omega(k^2 / \log^{1+\epsilon} k)$ -competitive [Ramesh, 1993]

New result

- randomized $O(k^2)$ -competitive
- implies same bound for *equivalent* problem of chasing sets of size $\leq k$ in a metric space (aka **metrical service systems**)

Model as Evolving Tree Game

Evolving tree game: Minimize movement while staying at leaf of binary tree that evolves in following ways:



Algorithm

Fork

Do (essentially) nothing

- new leaves at distance 0, so movement is free

Growth

$$x'_u = -\frac{2x_u \tilde{w}'_u}{\tilde{w}_u} + \frac{x_u + \delta_u}{\tilde{w}_u} \underbrace{(\lambda_{p(u)} - \lambda_u)}_{\text{Lagrange multipliers}}$$

$\tilde{w}_u \approx w_u = \text{edge length from } u \text{ to parent}(u)$

This is mirror descent with

- regularizer $\Phi_t(x) = \sum_u \tilde{w}_u (x_u + \delta_u) \log(x_u + \delta_u)$
- cost function $\tilde{w}'$ scaled by $\frac{2x_\ell}{x_\ell + \delta_\ell}$, where ℓ is growing leaf

Deletion

- Simulate growth $\tilde{w}_\ell \rightarrow \infty$ so that $x_\ell \rightarrow 0$.
- Then remove ℓ
- Merge edges at parent(ℓ)
- Update δ and $\tilde{w}$

Analysis

Use potential $P := 4kD - 2\Psi$ where

$$D := \sum_u \tilde{w}_u \left(2y_u \log \frac{1 + \delta_u}{x_u + \delta_u} + x_u \right) \quad \text{for } y = \text{offline}$$

$$\Psi := \sum_u \text{depth}(u) \tilde{w}_u x_u$$

and show

$$\sum_u w_u x'_u + \sum_u w'_u x_u + P' \leq O(k^2) \cdot \sum_u w'_u y_u$$

Why $O(k^2)$?

Similar to $O(\text{depth} \cdot \log \frac{1}{\delta_{\min}})$ for MTS. Here:

- $\text{depth} \leq \#\text{leaves} \leq k$
- Due to topology changes, need $\delta_{\min} = \frac{1}{2^{\text{depth}}} \geq \frac{1}{2^k}$

References

- W. R. Burley. Traversing layered graphs using the work function algorithm. *J. Algorithms*, 1996.
- A. Fiat, D. P. Foster, H. J. Karloff, Y. Rabani, Y. Ravid, and S. Vishwanathan. Competitive algorithms for layered graph traversal. In *FOCS*, 1991.
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